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RESEARCH

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Extra-pleural space on CT: common lesions and diagnostic approach

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Abstract

Background The extra-pleural space is challenging to view radiologically and is commonly overlooked. Therefore, it is important to have a comprehensive understanding of its anatomy in order to correctly classify lesions as either pleural or extra-pleural so as to choose the most appropriate treatment. CT with multi-planar reformatting allows the assessment of pathological conditions involving the extra-pleural space. The aim of this cross-sectional study was to assess the role of CT in the detection and evaluation of the extra-pleural space lesions and to classify them according to their CT appearance in order to reach a proper diagnosis and successful management plan.

Results This study was conducted on 131 patients who underwent CT scan of the chest for different chest complaints, and the detected extra-pleural space abnormalities were classified according to their CT appearance into three major groups which are fatty expansion and soft tissue stranding group that included 59 cases (45%), abnormal attenuation group that included 13 cases (10%), and soft tissue expansion group that included 59 cases (45%). Out of the 59 cases with increased fatty expansion and soft tissue stranding, 73% were of inflammatory cause (either pulmonary or pleural), 17% were of neoplastic cause, and 3 cases were caused by systemic conditions. The abnormal extra-pleural space attenuation group was classified into air-containing and blood-containing groups, and both were mainly caused by trauma. Soft tissue infiltration of extra-pleural space was classified into neoplastic (85%) and non-neoplastic (15%) lesions, while most of the non-neoplastic pathologies were caused by severe inflammatory processes. The axial images were mostly sufficient to detect extra-pleural space lesions.

Conclusions CT has a crucial role in the assessment of extra-pleural space and the different lesions involving it. Those lesions can be classified according to their CT appearance in order to reach a proper diagnosis and as a result aid in better management of these pathologies.

Keywords Extra-pleural space, CT, MPR

Background

The extra-pleural space (EPS) is a potential space that lies between the parietal pleura and the ribs' inner surface. It contains fat, connective tissue, lymph nodes, vessels, the endo-thoracic fascia, and the innermost intercostal muscle [1].

The EPS is challenging to view radiologically and is commonly overlooked during thoracic multi-detector computed tomography (MDCT) interpretation. Therefore, it is important to have a comprehensive understanding of the anatomy of the intra-pleural and extra-pleural spaces in order to correctly classify soft tissue lesions as either pleural or extra-pleural on thoracic imaging so as to choose the most appropriate treatment. CT aids in separating extra-pleural from pleural lesions by eliminating structure overlap [2].

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CT is considered the favorable imaging modality for evaluating the EPS and the pathologic conditions affecting this space. The mediastinal window allows for the assessment of EPS lesions with high spatial resolution, and multi-planar (sagittal, coronal, and oblique) reformatting may enhance the assessment of these lesions [3].

Most of the recent articles have classified the EPS pathologies according to their composition into three main categories. The first category is the extra-pleural fat expansion and soft tissue stranding which is caused by inflammatory conditions (either lung or pleural), neoplasms (intrathoracic peripheral malignancies, e.g., malignant pleural mesothelioma, metastases, and bronchogenic carcinoma), systemic conditions (obese people and chronic glucocorticoids users), and mimicking conditions which simulate EPS lesions specially those causing fatty expansion (e.g., Bochdalek's hernia and extra-pleural lipoma) [2–6].

The second category is the abnormal attenuation of EPS which is caused by blood-containing lesions (extra-pleural hematoma) or air-containing lesions (extra-pleural air) [4, 7, 8].

While the third category is the soft tissue expansion of the EPS caused by either non-neoplastic conditions as extramedullary hematopoiesis, amyloidosis, intrathoracic splenosis, and extra-pleural pneumolysis (plombage), or neoplastic conditions as malignant pleural mesothelioma, bronchogenic carcinoma, pleural metastasis, neurogenic tumors (along the path of the vagus, phrenic, recurrent laryngeal, or intercostal nerves), myeloproliferative neoplasms (lymphoma and leukemia), multiple myeloma (with extramedullary spread), and chest wall neoplasms [2–5, 9–13].

The aim of this study was to assess the role of CT in the detection and evaluation of the EPS lesions and to classify them according to their CT appearance in order to reach a proper diagnosis and successful management plan.

Methods

Study population

This cross-sectional prospective study received the approval of the ethical and scientific committees in our institute. Written informed consent was obtained from all patients.

This study initially involved 132 patients; 1 patient was excluded because of severe motion artifacts which precluded proper CT interpretation. Thereby, a total of 131 patients were included.

The study was conducted in our radiology department in the period from January 2021 to April 2022, where the patients were referred from the outpatient's clinic and chest department for CT chest. According to the clinician request, 107 (82%) patients underwent non-contrast CT chest study, while 24 (18%) patients underwent post-intravenous contrast CT study.

Inclusion criteria

Patients with suspected extra-pleural pathology in CT study.

Exclusion criteria

1. Patients with relative contraindication to perform CT study, e.g., pregnant, renal impairment, or history of hypersensitivity reaction to contrast material if contrast administration was required.
2. Studies with technical errors, e.g., motion artifact.

Methodology

All patients underwent thorough history taking and clinical examination by the chest physician. CT scan of the chest was done on all patients.

CT chest technique

CT scan of the chest was performed on all patients using Siemens SOMATOM Scope, Germany (CTAWP92544) 16-channel MDCT. CT scan parameters were as follows: slice thickness 1.5 mm, pitch 1.5 mm, gantry tilt 0, kilovolt (KV) 120, milliamperes (mAs) 25, rotation time 0.5 s, and total exposure time about 10 s.

The scans were performed in the supine position in full inspiration and covered the whole thorax from the root of the neck to the below diaphragm. Axial and reformatted sagittal and coronal images were taken in both lung [window width about 1600, window level about -600 Hounsfield unit (HU)] and mediastinal [window width about 400 and window level about 40 HU] windows.

In the case of intravenous contrast administrations, the dose was 1 ml/kg of non-ionic iodinated contrast (Omnipaque 350).

Interpretation of the CT studies

1. Pulmonary, pleural, or chest wall pathology was identified.
2. Evaluation of the EPS lesions regarding their location, size, extent, and CT density was done in the mediastinal window [window width about 400 and window level about 40 HU].
3. The nature of the lesion was assessed, i.e., neoplastic, inflammatory, post-traumatic, or others.
4. The presence of specific signs indicative of EPS pathology, e.g., extra-pleural fat sign, and convexity of extra-pleural hematoma sign were also assessed.

Table 1 Major presenting chest complaints of the patients in our study

Clinical presentation	Dyspnea	Cough	Chest pain	Chest swelling
Number	110	72	55	2
Percentages	84%	55%	42%	2%

Table 2 EPS lesions number and percentage in each group of EPS pathology

EPS abnormality	No. of patients	Percentage (%)
Fatty expansion and soft tissue stranding	59	45
Abnormal attenuation	13	10
Soft tissue expansion	59	45

Histopathological evaluation

Histopathological evaluation was done in 62 patients (47%) either by ultrasound or by CT-guided biopsy.

Statistical analysis

On the program known as Statistical Package of Social Science Software version 25, data were entered and statistical analysis was performed (IBM SPSS Statistics for Windows, Version 25.0. Armonk, NY: IBM Corp.). Frequency and percentage were used to show the data for quantitative variables.

Results

This study was conducted on 131 patients: 89 males (68%) and 42 females (32%), with an age range from 19 to 89 years (mean age of 57 years).

The referred patients were primarily suffering from either dyspnea or cough, as detailed in Table 1.

The detected EPS abnormalities were classified according to their CT appearance into three major groups as illustrated in Table 2.

Fatty expansion and soft tissue stranding group

The fatty expansions and soft tissue stranding group were further divided into inflammatory processes (either pulmonary or pleural), neoplastic processes, systemic conditions, and mimickers conditions as presented in Table 3.

The inflammatory conditions that caused extra-pleural space fatty expansion and soft tissue stranding are recorded in Table 4.

The 10 neoplastic conditions that caused EPS fatty expansion and soft tissue stranding were 5 (50%) cases of bronchogenic carcinoma, 3 (30%) cases of malignant pleural mesothelioma, and 2 (20%) cases of metastatic deposits.

The 3 cases of systemic conditions which caused fatty expansion were caused by chronic corticosteroid use (Fig. 1).

Table 3 Number and percentage of patients in the fatty expansion and soft tissue stranding group

Pathology	No. of patients	Percentage (%)
Inflammation	43	73
Neoplasms	10	17
Systemic conditions	3	5
Mimickers	3	5

Table 4 Number and percentage of patients with inflammatory conditions leading to fatty expansion and soft tissue stranding of the EPS

Pathology	No. of patients	Percentage (%)
Pneumonia	26	61
Asbestos exposure disease	10	23
Empyema	3	7
Lung fibrosis	3	7
Ruptured aortic aneurysm	1	2



Fig. 1 50-year-old male patient with known history of chronic corticosteroids administration presented with shortness of breath and chest tightness. Axial CT mediastinal window image showing bilateral rather symmetrical fatty expansion of the EPS (arrows)



Fig. 2 54-year-old male patient complaining of dyspnea. Axial CT mediastinal window image showing an incidentally discovered extra-pleural lipoma

The 3 mimicking conditions were 2 (66%) cases of extra-pleural lipoma (Fig. 2) and 1 (33%) case of diaphragmatic hernia.

Abnormal EPS attenuation group

The second group which is the abnormal EPS attenuation group included 5 (38%) cases of air attenuation lesions (Fig. 3A) and 8 (62%) cases of blood attenuation lesions (Fig. 3B).

Seven (87.5%) of the blood-containing lesions (extra-pleural hematomas) were post-traumatic and one (12.5%) case was iatrogenic (post-chest tube insertion).

Three (48%) out of those eight cases showed biconvex appearance suggesting arterial injury, while five (52%) cases showed linear/nodular inner outline suggesting slow flow (i.e., venous) injury. Also, five (62.5%) patients showed underlying ribs fractures.

However, four (80%) cases of the air-containing lesions were post-traumatic and one (20%) case was iatrogenic (post-central venous line insertion).

EPS infiltration by soft tissue attenuation group

The last group of cases where the EPS was infiltrated by soft tissue attenuation lesions included 47 (85%) neoplastic and 12 (15%) non-neoplastic lesions.

The detected neoplastic lesions infiltrating the EPS in our study are reported in Table 5.

The non-neoplastic lesions expanding the EPS with soft tissue were 9 (75%) cases of inflammatory nature and 3 (25%) cases of extramedullary hematopoiesis (Fig. 4).

Overall, the EPS lesions in our study were 44.3% of inflammatory cause, 43.5% were of neoplastic cause and 9.9% were post-traumatic.

The site of the underlying pathology that involved the EPS was either caused by parenchymal lung (43.5%), pleural (38.9%), or chest wall (17.6%) pathology.

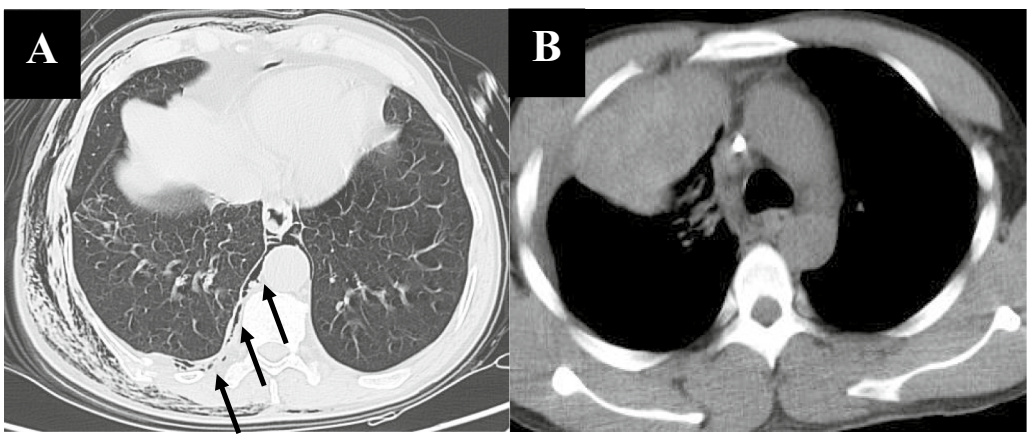


Fig. 3 Abnormal attenuation of the EPS. **A** 41-year-old male patient presented with poly-trauma secondary to road traffic accident. Axial CT mediastinal window image showing rib fracture, surgical emphysema, and right lower lobe EPS air (very fine weblike internal septations characteristic of extra-pleural air) (arrows). **B** 39-year-old male patient presented with sudden dyspnea and shortness of breath after central venous line insertion. Axial CT mediastinal window image showing a relatively hyper-dense biconvex-shaped EPS hematoma post-central venous line insertion, i.e., iatrogenic (convexity of extra-pleural hematoma sign)

Table 5 Number and percentage of patients with neoplastic conditions leading to soft tissue infiltration of the EPS

Pathology	No. of patients	Percentage (%)
Metastasis	17	36
Malignant pleural mesothelioma	14	30
Bronchogenic carcinoma	10	21
Neurogenic tumor	2	4
Chondrosarcoma	1	2
Round cell tumor	1	2
Lymphoma	1	2
Leiomyosarcoma	1	2



Fig. 4 24-year-old female patient with known history of thalassemia presented with chest pain. Axial CT mediastinal window image of the upper chest showing paravertebral calcified extramedullary hematopoiesis involving the EPS

Table 6 Detection of lesions in different planes

Plane	Number (%)
Axial	131 (100%)
Coronal	102 (78%)
Sagittal	29 (22%)

Most of the identified EPS lesions were unilateral in as seen 91 (83.2%) cases with only 22 (16.8%) cases showing bilateral lesions (e.g., in metastatic pathology or diffuse inflammatory process).

Contrast administration was done in 56 (42.7%) cases; however, it did not add further information regarding EPS affection.

The axial images were mostly sufficient to detect EPS lesions as emphasized in Table 6.

Discussion

Due to a relative lack of awareness of the anatomy and imaging appearances of the disorders affecting the EPS, it is frequently ignored in imaging [2].

As the different pathologies involving the EPS may require different treatment approaches, so, the proper understanding of the EPS is essential for radiologists [3]. Additionally, the involvement of EPS may have an impact on the staging as well as the treatment plan of chest neoplasms [3].

CT has good soft tissue contrast and spatial resolution, so it is the optimal imaging technique to be used in evaluating the EPS [4].

The purpose of this study was to identify the different pathologies that may involve the EPS and to classify them according to their CT appearance in order to reach a proper diagnosis.

This study is a relative pioneer study of the EPS lesions, thus surfing the Internet for the literature of similar subject was relatively a hard task where only a limited number of articles were found.

Also, in reviewing the literature, no statistical data could be obtained due to the small size and inhomogeneity of the samples obtained; as Valent et al. mentioned the largest sample was about 29 cases [4].

In order to classify the type of lesions involving the EPS, we used the radiological approach according to the appearance of lesions in CT as in Valente et al. study [4].

This study was conducted on 131 cases with EPS lesions seen by CT; they were classified into three major groups which are the fatty expansion and soft tissue stranding group that included 59 (45%) cases, abnormal attenuation group that included 13 (10%) cases, and the last group which is the soft tissue expansion group that included (45%) 59 cases.

Santamarina et al. used a different approach and classified the EPS lesions according to their etiology to inflammatory, traumatic, neoplastic, and miscellaneous causes [2].

Another approach used by Maheshwarappa et al. classified the EPS lesions according to their focality, i.e., focal or diffuse [3].

In this study, the fatty expansion and soft tissue stranding group included the causes that led to increased thickness and attenuation of this intercostal stripe which is known as extra-pleural fat sign (Fig. 5); on CT this appears as an inward movement of an extra-pleural fat stripe [14].

Maheshwarappa et al. declared that increased extrapleural fat is usually seen in benign conditions, and less commonly it can be noted in certain malignancies which result in lung scarring or causes irritation of the pleura. This agreed with our study as out of the 59 cases

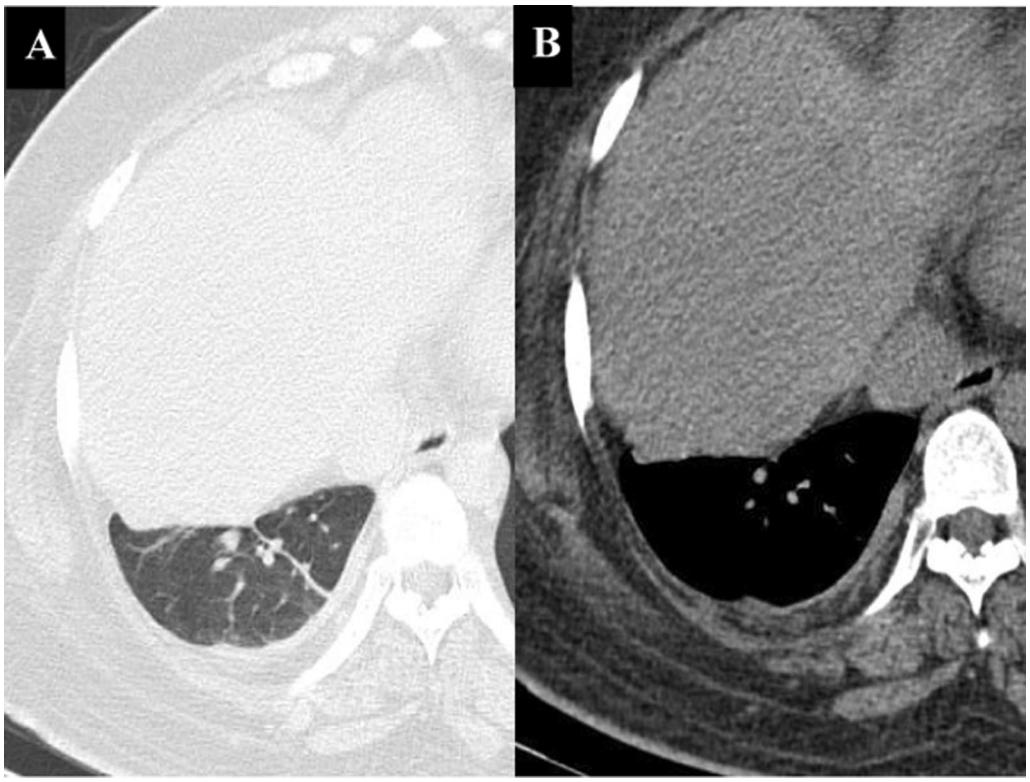


Fig. 5 37-year-old male patient with past history of right lower lobe pneumonia 1 month before presented with dyspnea and chest tightness. Axial CT cuts in **(A)** lung window and **(B)** mediastinal window showing post-inflammatory scarring and expansion of the extra-pleural fat (extra-pleural fat sign)

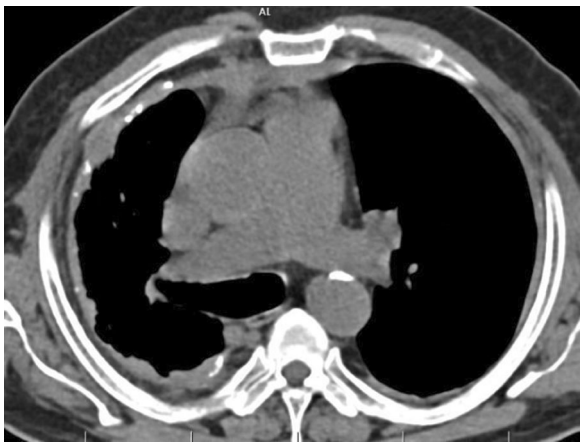


Fig. 6 69-year-old male patient presented with cough, dyspnea and chest pain. Axial CT mediastinal window images showing extra-pleural fatty expansion and soft tissue stranding in a case of mesothelioma on top of asbestos exposure disease

with fatty expansion and soft tissue stranding, 73% were of inflammatory cause (either pulmonary or pleural) (Fig. 5), 17% were of neoplastic cause (Fig. 6), and only 3 cases were caused by systemic conditions [3].

The abnormal EPS attenuation group was classified into air-containing and blood-containing groups.

In accordance with Santamarina et al. and Maheshwarappa et al., abnormal accumulation of air in the EPS was mainly due to traumatic causes as seen in 75% of our cases [2, 3].

It is critical to distinguish between EPS air and pneumothorax since the management of these conditions is different. Extra-pleural air usually resolves spontaneously, although pneumothorax, if severe, may necessitate the insertion of an intercostal tube. The appearance of "weblike" linear opacities inside the air collection on CT is one of the radiological markers that aid in the localization of air to the EPS [2, 3].

Similarly, extra-pleural hematoma which represents the blood-containing group was mainly due to traumatic causes as seen in 87.5% of our cases.

This agreed with Santamarina et al. and Valente et al. who claimed that extra-pleural hematomas were most commonly due to injury to intercostal or internal mammary arteries or veins caused by blunt or penetrating trauma [2, 4].

In our study, 3 patients with extra-pleural hematoma showed biconvex border (Fig. 3B); all of them showed

underlying ribs fractures (suggesting penetrating injury), while 5 cases showed nonconvex border; only two of them showed underlying ribs fractures (suggesting blunt trauma).

This was consistent with Chung et al. who classified extra-pleural hematoma into 2 types with different prognosis and therapeutic implications: biconvex and nonconvex. Large hematomas are typically biconvex, indicating arterial rather than venous bleeding, and may necessitate surgery or trans-catheter arterial embolization. However, nonconvex extra-pleural hematoma is commonly caused by venous (low-pressure) bleeding, usually smaller in size and can be treated conservatively [15].



Fig. 7 50-year-old male patient with history of persistent cough and hemoptysis. Axial CT mediastinal window image of the right upper lung lobe showing a pleural-based mass lesion invading the EPS (pathologically proved bronchogenic carcinoma)

In this study, we classified soft tissue infiltration of EPS into neoplastic or non-neoplastic processes; 47 (85%) of the cases were of neoplastic nature, while only 12 (15%) cases were of non-neoplastic nature.

The malignancies that involved the EPS in this study were 17 cases (36%) of metastatic nature, 14 cases (30%) of malignant pleural mesothelioma, 10 cases (21%) of bronchogenic carcinoma (Fig. 7), and 6 cases (13%) of other chest wall neoplasms originating from the EPS including nerve sheath tumors, round cell tumor, lymphoma, chondrosarcoma, and leiomyosarcoma (Fig. 8).

Maheshwarappa et al. stated that invasion of the EPS is pivotal in staging of most of lung and pleural tumors, and it significantly affects the treatment plan of the patient, e.g., invasion of the EPS by a primary non-small cell lung cancer of any size is categorized as stage T3 disease [3].

In our study, the inflammatory conditions that caused EPS soft tissue expansion were severe inflammatory process: 2 cases of empyema necessitans (Fig. 9), 6 cases of advanced asbestos exposure disease and 2 of post-inflammatory fibrothorax, in addition to 2 cases of advanced extramedullary hematopoiesis. This conformed to Maheshwarappa et al. who declared that the soft tissue expansion of EPS by non-neoplastic process indicates the severity and the bad prognosis of the disease [3].

All the lesions in our study were well identified and evaluated in axial CT cuts with mediastinal window, 102 (78%) of the lesions were identified in coronal images and only 29 (22%) of the lesions could be identified in sagittal images, indicating that axial images were sufficient for detection and diagnosis of EPS lesions, and this agreed

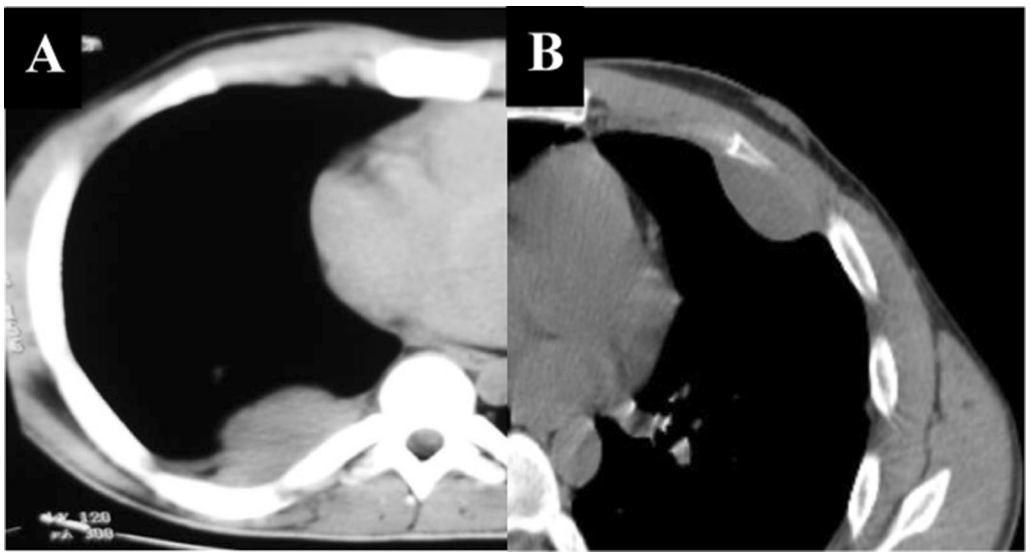


Fig. 8 **(A)** 22-year-old male patient presented with chest pain and dyspnea. Axial CT mediastinal window image showing involvement of the EPS with a mass lesion (pathologically proved round cell tumor) displacing the extra-pleural fat stripe. **(B)** 45-year-old male patient complaining of chest wall swelling and pain. Axial CT mediastinal window image of the lingular segment showing EPS mass (pathologically proved nerve sheath tumor)

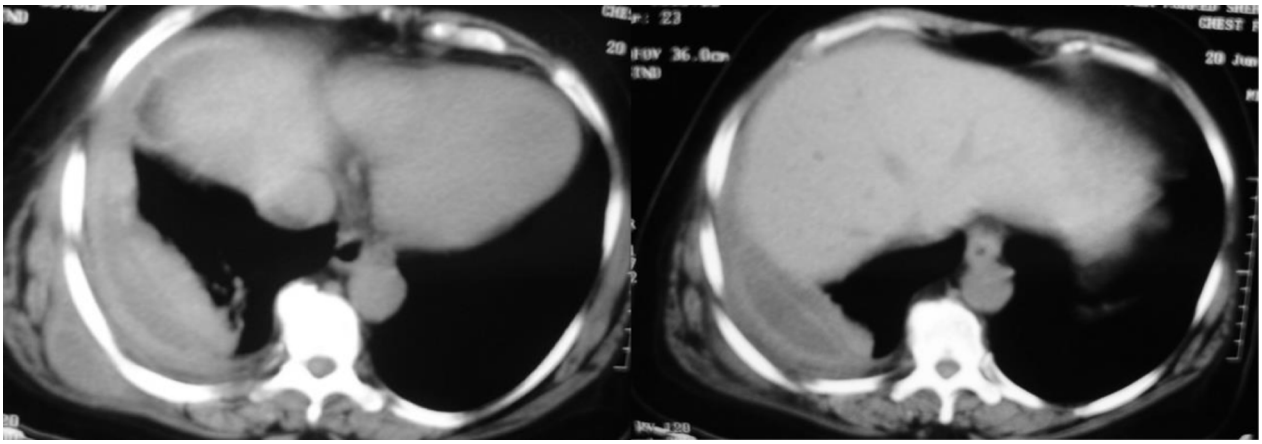


Fig. 9 45-year-old female patient presented with fever, chest pain and cough. Axial CT mediastinal window images showing a case of right basal empyema (split pleura sign), empyema necessitans and soft tissue expansion of the extra-pleural space

with Valente et al., who claimed that although the current generation of multi-detector CT scanners generates data sets with high spatial and contrast resolution allowing detailed multi-planar reconstructions in post-processing evaluation, the axial sections remain the mainstay of EPS lesions interpretation [4].

Our study limitation is the relatively small number of cases, so we recommend further research on a larger population to confirm our results and obtain statistical data.

Conclusions

CT has a crucial role in the assessment of EPS and the different lesions involving it. Those lesions can be classified according to their CT appearance in order to reach a proper diagnosis and as a result aiding in better management of these pathologies.

Abbreviations

EPS	Extra-pleural space
HU	Hounsfield unit
KV	Kilovolt
mAs	Milliampere
MDCT	Multi-detector computed tomography

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Not applicable.

Author contributions
YYS, SFT, and MSE reviewed the images. YYS, MAK, YHE, and MRA analyzed and interpreted the patient data. SFT wrote the manuscript and YYS reviewed it. All authors have read and approved the manuscript.

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Availability of data and materials
The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent for participate
Approval of the ethical committee of the 'Radiology department, Faculty of Medicine, Cairo University' was granted before conducting this prospective study; Reference number: not applicable; local institutional review board approval was granted before conducting this cross-sectional study, and written informed consent was obtained from all patients.

Consent for publication
All patients included in this research gave written informed consent to publish the data contained within this study. If the patients were less than 16 years old, deceased, or unconscious when consent for publication was requested, written informed consent for the publication of these data was given by their parents or legal guardians.

Competing interests
The authors declare that they have no competing interests.

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